Models of Systems

Module 2 - ABE 721: Agricultural Systems Modelling and Simulation

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Lecture Outline

1. 1. Introduction and Terminologies
2. 2. Classification and Taxonomy of Models
3. 3. The Modelling Process
4. 4. Model Formulation Examples

The Philosophy of Modelling

George E. P. Box (1976)

"All models are wrong, but some are useful."

Why do we model?

- To simplify infinitely complex, chaotic agricultural realities (soil physics, plant physiology, machinery dynamics).
- To yield predictive, explanatory, or normative (optimizing) insights.
- **The Art of Abstraction:** Knowing exactly which variables to ignore (e.g., ignoring aerodynamic drag when calculating deep tillage draft force).

Core Systems Engineering Terminologies

- **System Boundary:** The conceptual/mathematical line separating the system from its environment.
- **State Variables ($\mathbf{x}$):** Variables that perfectly describe the condition of the system at a moment in time (e.g., soil moisture $\theta(t)$).
- **Parameters ($\mathbf{p}$):** Constants that define the physical characteristics of equations (e.g., saturated hydraulic conductivity K_s).
- **Exogenous Variables ($\mathbf{u}$):** Environmental inputs that force the system, entirely independent of the system's state (e.g., solar radiation, rainfall).

Mathematical State-Space Representation

Continuous-time dynamic engineering models are generally represented by coupled Ordinary Differential Equations (ODEs):

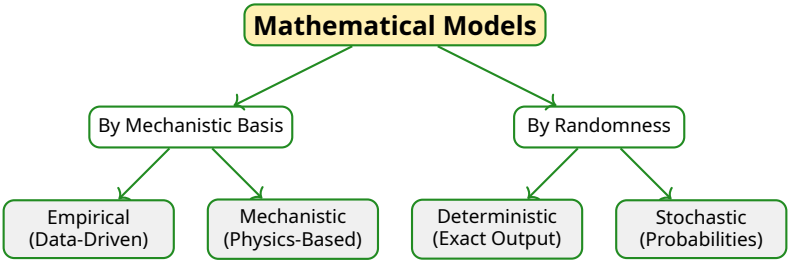
$$\frac{d\mathbf{x}(t)}{dt} = f(\mathbf{x}(t), \mathbf{u}(t), \mathbf{p}, t) \tag{1}$$

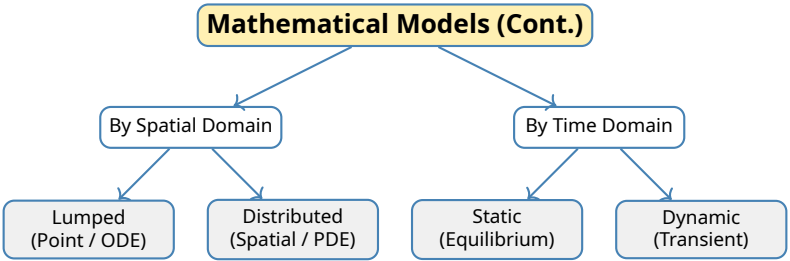
If the system is *Linear and Time-Invariant (LTI)*, this simplifies beautifully into standard matrix algebra:

$$\dot{\mathbf{x}}(t) = \mathbf{A}\mathbf{x}(t) + \mathbf{B}\mathbf{u}(t) \quad \text{(State Equation)} \tag{2}$$

$$\mathbf{y}(t) = \mathbf{C}\mathbf{x}(t) + \mathbf{D}\mathbf{u}(t) \quad \text{(Output Equation)} \tag{3}$$

Taxonomy of Agricultural Models





Empirical vs. Mechanistic Models

Empirical (Black-Box)

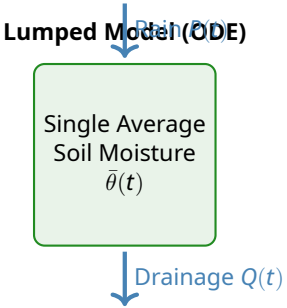
- Derived from observational data (Regression, Neural Networks).
- E.g., Yield = $f(\text{Rain}, \text{Fertilizer})$.
- *Pros:* Fast, easy if data is huge.
- *Cons:* Fails completely if extrapolated to novel scenarios (like climate change).

Mechanistic (White-Box)

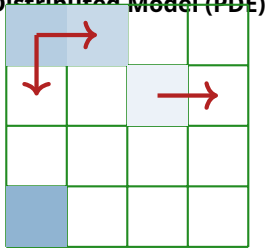
- Built on physics/biology (Conservation of Mass, Thermodynamics).
- Explains *why* the system acts as it does.
- *Pros:* Highly robust under extreme conditions.
- *Cons:* Very data-hungry to calibrate physiological parameters.

Spatial Domain: Lumped vs. Distributed

Lumped Model (ODE)

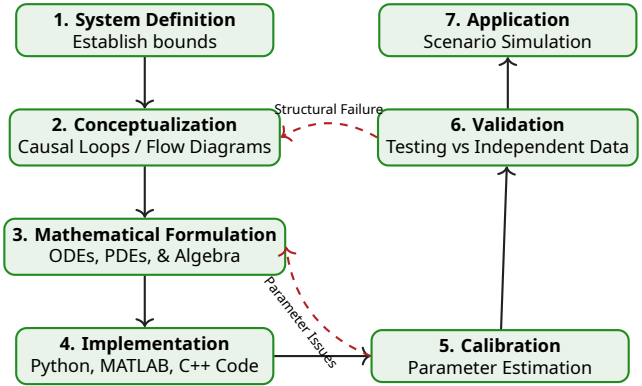


Distributed Model (PDE)



Spatially variable $\theta(x, y, t)$
interacting via fluxes $\mathbf{q}$

The Iterative Systems Modelling Process



Formulation Example: Soil Water Dynamics

The Richards Equation models how water moves vertically (z) through unsaturated soil over time (t).

It combines the **Continuity Equation** (Conservation of Mass) with **Darcy's Law** for unsaturated flow:

$$q = -K(\theta)\left(\frac{\partial h}{\partial z} - 1\right)$$

1D Mixed-form Richards Equation

$$\frac{\partial \theta}{\partial t} = \frac{\partial}{\partial z}\left[K(\theta)\left(\frac{\partial h}{\partial z} - 1\right)\right] - S(z,t)$$

Note: $K(\theta)$ is hydraulic conductivity (highly non-linear), h is matric suction, and S is root water uptake (sink).

Formulation Example: Greenhouse Thermodynamics

Predicting internal air temperature (T_{in}) using the **First Law of Thermodynamics** (Conservation of Energy).

Rate of change of thermal energy = Sum of heat fluxes.

$$C_{air}\frac{dT_{in}}{dt} = Q_{solar} + Q_{heater} - Q_{cond} - Q_{vent} - Q_{ET}$$

Expanded ODE Formulation

$$(V\rho_a c_p)\frac{dT_{in}}{dt} = \tau A_g I_s + Q_{heater} - UA_c(T_{in} - T_{out}) - \dot{m}c_p(T_{in} - T_{out}) - \lambda E_T$$

Used to design automated PID controllers for the Q_{heater} and ventilation fans ($\dot{m}$).

Summary of Module 2

- **Modelling Philosophy:** Abstraction of reality to solve specific problems.
- **Taxonomy:** We distinguish models based on mechanics (Empirical vs. Mechanistic) and dimensions (Lumped vs. Distributed).
- **Process:** Modelling is highly iterative. Failure at validation (Step 6) requires returning to conceptualization (Step 2).
- **Formulation:** Translating physics (Mass, Energy, Momentum) into rigorous Ordinary and Partial Differential Equations (ODEs/PDEs).

End of Module 2

Questions & Discussion